WEEK 5

1. A data automation team is creating a utility module that performs multiple independent calculations based on user inputs. Write a Python program that accepts required inputs from the user and defines separate functions to perform:
   - Find the largest of three numbers (positional arguments).
   - Compute the volume based on user's chosen shape: – cylinder (r, h), cube (a), or rectangular box (l, w, h).
   - Compute the area of a rectangle.
   - Compute the circumference of a circle.
   - Exchange the values of two variables.
   - Find the distance between two points (x1, y1) and (x2, y2) using math.dist(). (Use switch case to execute the required function)

```
import math
def largest(a,b,c):
    return max(a,b,c)

def volume():
    shape=input("Enter the shape : ")
    if shape=="Cuboid":
       l=float(input("Enter the length: "))
       b=float(input("Enter the breadth: "))
       h=float(input("Enter the height: "))
       return l*b*h
    elif shape=="Cube":
        side=int(input("Enter the side: "))
        return side**3
    elif shape=="Cylinder":
        h=float(input("Enter the height : "))
        r=float(input("Enter the radius: "))
        return  math.pi*(r**2)*h

def areaofrectangle(a,b):
    return a*b

def circumofcircle(r):
    return 2*math.pi*r

def swapping(a,b):
    return b,a

def distbtwpoints(a,b,c,d):
    d=math.sqrt((c-a)**2 + (d-b)**2)
    return d
```

```
print("\n-----Data Automation Module------\n")
print("\n.Largest \n2.Volume \n3.Area of rectangle \n4.Circumference \n5.Swap \n6.Distance between 2 points\n")
choice=input("Enter the choice (1-6): ")

match choice:
    case '1':
        x=int(input("Enter the first number: "))
        y=int(input("Enter the second number: "))
        z=int(input("Enter the third number: "))
        print("Largest: ",largest(x,y,z))

    case '2':
        print("volume: ",volume())

    case '3':
        l=int(input("Enter the length: "))
        b=int(input("Enter the breadth: "))
        print("Area: ",areaofrectangle(l,b))

    case '4':
        r=float(input("Enter the radius:"))
        print("Circumference: ",circuumofcircle(r))

    case '5':
        a=int(input("Enter the first value: "))
        b=int(input("Enter the second value: "))
        print("Swapped: ",swapping(a,b))
```

```
  case '6':
       a=int(input("Enter the first value of coordinate 1: "))
       b=int(input("Enter the second value of coordinate 1: "))
       c=int(input("Enter the first value of coordinate 2: "))
       d=int(input("Enter the second value of coordinate 2: "))
       print("distance: ",distbtwpoints(a,b,c,d))

  case _:
      print("Invalid Choice" )
```

```
-----Data Automation Module------


.Largest
2.Volume
3.Area of rectangle
4.Circumference
5.Swap
6.Distance between 2 points

Enter the choice (1-6): 2
Enter the shape : Cube
Enter the side: 4
volume:  64
```

2. A billing system receives an unknown count of numeric entries per invoice. Write a Python function that accepts arbitrary integers and returns their sum. (Hint: use *args)

```
def find_sum(*args):
    return sum(args)

print(find_sum())
print(find_sum(5,10))
```

```
0
15
```

3. A text-filtering tool needs to remove noisy characters occurring at odd positions. Write a Python program that takes a string and removes all characters at even index values.

   Example 1: • Input: "H@e#l$l" • Output: "el"

```
string=input("enter a string:")
result=""

for i in range(len(string)):
    if string[i].isalpha() or i%2==0:
        result+=string[i]

print(result)
leftstr=""

for i in range(len(result)):
    if i%2!=0:
        leftstr+=result[i]

print("resut after removing even index characters:",leftstr)
```

```
enter a string:hello!!98
hello!8
resut after removing even index characters: el!
```

4. Write a Python script that takes input from the user and displays that input back in upper and lower cases.

```
string=input("enter the string:")
print("upper case:",string.upper())
print("lower case:",string.lower())
```

```
enter the string:HELLO WORLD
upper case: HELLO WORLD
lower case: hello world
```

5. A simple analytics module must calculate keyword frequency in user feedback. Write a Python program to count occurrences of each word in a sentence.

   Example 1: • Input: "Hello hello world" • Output: {'hello': 2, 'world': 1}

```
s=input("enter sentence:")
words=s.lower().split()
word_count={}
for word in words :
    if word in word_count:
        word_count[word]+=1
    else:
        word_count[word]=1

print(word_count)
```

```
enter sentence:hello world hello programming
{'hello': 2, 'world': 1, 'programming': 1}
```

6. A document cleanup tool needs to normalize text copied from notes. Write a Python program to remove indentation from each line of a multi-line text.

```
print("enter multi-line text(press Enter twice to finish):")

lines =[]
while True:
    line =input()
    if line==="":
        break
    lines.append(line)

print("\ntext after removing indentation:")

for line in lines:
    print(line.lstrip())
```

```
enter multi-line text(press Enter twice to finish):
hello
      world


text after removing indentation:
hello
world
```

7. Write a function that reverses a string. The input string is given as an array of characters s. You must do this by modifying the input array in place with O(1) extra memory.

```
def reverse_string(chars):
    left =0
    right=len(chars)-1

    while left<right:
        chars[left],chars[right]= chars[right],chars[left]
        left+=1
        right-=1

s= input("Enter a string:")

char_list=list(s)

reverse_string(char_list)

print("Reversed string:","".join(char_list))
```

```
Enter a string:hello world
Reversed string: dlrow olleh
```

8. Given a string s, reverse the order of characters in each word within a sentence while still preserving whitespace and initial word order.

```
def reverse(sentence):
    result=[]
    word=""

    for ch in sentence:
        if ch.isspace():
            result.append(word[::-1])
            result.append(ch)
            word=""
        else:
            word+=ch

    result.append(word[::-1])
    return "".join(result)
s=input("enter a sentence:")
output=reverse(s)
print("new sentence is:",output)
```

```
enter a sentence:python is a programming language
new sentence is: nohtyp si a gnimmargorp egaugnal
```

9. A parser converts messages into list-based tokens. Write a Python program to convert a string into a list of characters.

Example 1: • Input: "Hi 5" • Output: ['H','i',' ','5']

```
s=input("Enter the string: ")
character=[]
for ch in s:
    character.append(ch)

print(character)
```

```
Enter the string: hello
['h', 'e', 'l', 'l', 'o']
```

10. Write a Python program to count and display all vowels present in the text. Also display consonants and its count.

```
s=input("Enter the string: ")
v_count=0
c_count=0

for ch in s:
    if ch=='a' or ch=='e' or ch=='i' or ch=='o' or ch=='u'
       or ch=='A' or ch=='E' or ch=='I' or ch=='O' or ch=='U':
        v_count+=1
    else:
        c_count+=1

print("Number of vowels: ",v_count)
print("Number of consonants: ",c_count)
```

```
Enter the string: hello world
Number of vowels:  3
Number of consonants:  8
```